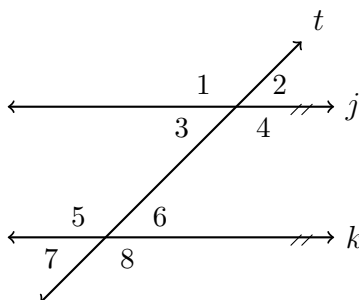


Naming the Angle Pairs a Transversal Makes

- In every figure below, the double arrowheads on two lines mean those lines are **parallel**. Where the marks are missing, you may not assume it.
- **Interior** angles lie between the two lines; **exterior** angles lie outside them. “Alternate” means opposite sides of the transversal; “same-side” means the same side.
- Name the pair first, then write what the name gives you — congruent (equal measures) or supplementary (measures adding to 180) — and only then compute.

1. In the figure, $j \parallel k$ and t is a transversal.

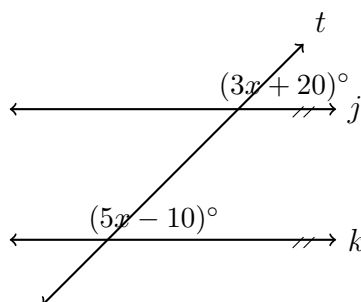


- (a) Name the relationship between $\angle 3$ and $\angle 6$, and the relationship between $\angle 4$ and $\angle 6$. For each, say whether the two angles are congruent or supplementary.

- (b) Given that $m\angle 4 = 118$ degrees, find $m\angle 6$.

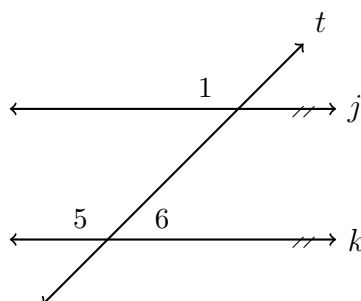
Answer: _____ degrees

2. $j \parallel k$. The two marked angles are corresponding angles. Find x .



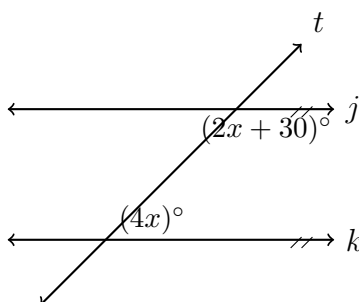
Answer: _____

3. $j \parallel k$ and $m\angle 1 = 105$ degrees. Find $m\angle 6$. (Work in two steps: first relate $\angle 1$ to $\angle 5$, then relate $\angle 5$ to $\angle 6$.)



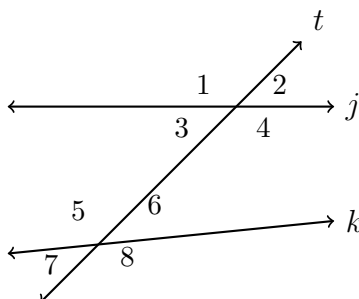
Answer: _____ degrees

4. $j \parallel k$. The two marked angles are same-side interior angles. Find x .



Answer: _____

5. In this figure the two lines carry *no* parallel marks, so they need not be parallel.

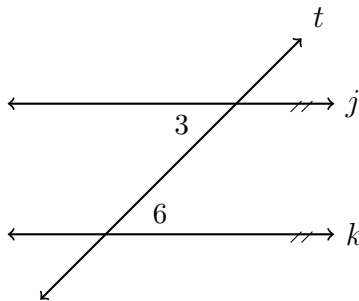


(a) Which angle-pair relationships still hold in this figure, and which ones need the lines to be parallel? Explain how you can tell.

(b) Given that $m\angle 2 = 74$ degrees, find $m\angle 4$.

Answer: _____ degrees

6. $j \parallel k$, and $\angle 3$ and $\angle 6$ are alternate interior angles with $m\angle 3 = (4x + 10)^\circ$ and $m\angle 6 = (6x - 20)^\circ$.



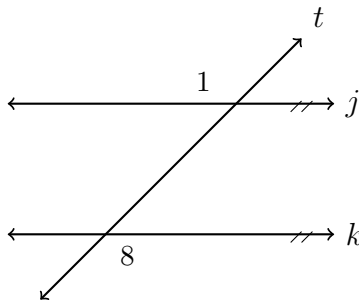
- (a) Find x .

Answer: _____

- (b) Find $m\angle 3$.

Answer: _____ degrees

7. $j \parallel k$, and $\angle 1$ and $\angle 8$ are alternate exterior angles with $m\angle 1 = (5x - 5)^\circ$ and $m\angle 8 = (3x + 31)^\circ$.



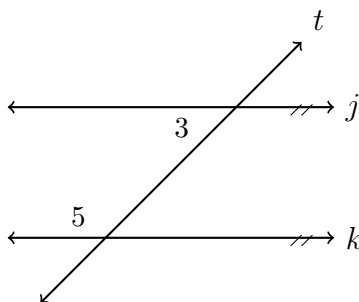
- (a) Find x .

Answer: _____

- (b) Find $m\angle 1$.

Answer: _____ degrees

8. $j \parallel k$, with $m\angle 3 = (4x + 6)^\circ$ and $m\angle 5 = (6x + 4)^\circ$.



- (a) Name the pair $\angle 3$ and $\angle 5$ form, and explain why that name means you must add the two expressions rather than set them equal.

- (b) Find x .

Answer: _____

- (c) Find $m\angle 5$.

Answer: _____ degrees